

# Summary for 2024-10-28-ApplsLinProg

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## 1 Summary

This Linear Programming lecture explored various applications and formulations of optimization problems. The core concept revolves around constructing and solving mathematical models to find optimal solutions under given constraints.

\* \*\*Production Planning:\*\* This application involved optimizing an ice cream manufacturer's yearly production plan to minimize costs associated with changing production levels and storing surplus inventory. The problem was formulated as a linear program, requiring linearization of absolute value terms in the objective function.

\* \*\*Sales Force Planning:\*\* This problem focused on minimizing labor costs by optimally allocating work between regular and temporary employees, given fluctuating monthly labor demands. The initial non-linear formulation was successfully transformed into a solvable linear program.

\* \*\*Portfolio Selection (Markowitz Model):\*\* This application involved maximizing portfolio return while considering risk. The initial formulation, a quadratic program due to the inclusion of variance, was discussed, along with its approximation as a linear program. Methods for estimating expected returns and variances from historical data were also presented.

\* \*\*Smallest Ball Problem:\*\* This geometric problem, finding the smallest ball containing a set of points, was formulated as a convex quadratic program. A theorem guaranteeing the existence and uniqueness of the solution was presented.

\* \*\*Maximum Weight Matching:\*\* This combinatorial optimization problem, represented as a bipartite graph, was formulated as an integer program to find the matching that maximizes the total weight.

\* \*\*Machine Scheduling:\*\* This problem involved assigning jobs to machines to minimize total completion time. It was formulated as an integer program with constraints ensuring each job is assigned to exactly one machine.

\* \*\*Knapsack Problem:\*\* This classic problem, selecting items to maximize value under a weight constraint, was presented as an integer program with binary decision variables.

\* \*\*Cutting Stock Problem:\*\* This problem, optimizing the cutting of large paper rolls into smaller ones to minimize waste, was formulated as an integer linear program. Various cutting patterns were considered, and constraints ensured customer order fulfillment.

\* \*\*Sparse Solutions of Linear Systems:\*\* This problem, finding sparse solutions to underdetermined linear systems, was explored in the context of signal recovery and data compression. The problem was formulated as an optimization problem involving minimizing the  $l_1$  norm of the solution vector, which can be transformed into a linear program.